

Automated QA, Review & Honeypots

Guidelines for assessing annotation quality in CVAT automatically

In CVAT, it’s possible to evaluate the quality of annotation through the creation of a validation subset of images. To estimate the task quality, CVAT compares all other jobs in the task against the established **Ground truth** job, and calculates annotation quality based on this comparison.

- Note** that quality estimation only supports 2d tasks. It supports all the annotation types except 2d cuboids.
- Note** that quality estimation is currently available for tasks and jobs. Quality estimation in projects is not supported.

CVAT has the following features for automated quality control of annotations:

- Validation set configuration for a task
- Job validation on job finish (“[Immediate feedback](#)”)
- Review mode for problems found
- Quality analytics

Basics

There are several approaches to quality estimation used in the industry. In CVAT, we can use a method known as Ground Truth or Honeypots. The method assumes there are Ground Truth annotations for images in the dataset. This method is statistical, which means that we can use only a small portion of the whole dataset to estimate quality on the full dataset, so we don’t need to annotate the whole dataset twice. Here we assume that the images in the dataset are similar (represent the same task).

We will call the validation portion of the whole dataset (or a task in CVAT) a validation set. In practice, it is typically expected that annotations in the validation set are carefully validated and curated. It means that they are more expensive - creating them might require expert annotators or just several iterations of annotation and validation. It means that it’s desirable to keep the validation set small enough. At the same time, it must be representative enough to provide reliable estimations. To achieve this, it’s advised that the validation set images are sampled randomly and independently from the full dataset. That is, for the quality assurance to function correctly, the validation set must have some portion of the task frames, and the frames must be chosen randomly.

Depending on the dataset size, data variance, and task complexity, **5-15% of the data is typically good enough** for quality estimation, while keeping extra annotation overhead for the Ground Truth acceptable.

For example, in a typical **task with 2000 frames**, selecting **just 5%**, which is 100 extra frames to annotate, **is enough** to estimate the annotation quality. If the task contains **only 30 frames**, it’s advisable to select **8-10 frames**, which is **about 30%**. It is more than 15%, but in the case of smaller datasets, we need more samples to estimate quality reliably, as data variance is higher.

Ground truth jobs

A **Ground Truth job** (GT job) is a way to represent the validation set in a CVAT task. This job is similar to regular annotation jobs - you can edit the annotations manually, use auto-annotation features, and import annotations in this job. There can be no more than 1 Ground Truth job in a task.

To enable quality estimation in a task, you need to create a Ground truth job in the task, annotate it, switch the job stage to `acceptance` , and set the job state to `completed` . Once the Ground Truth job is configured, CVAT will start using this job for quality estimation.

Read more about Ground Truth management [here](#).

Configuring quality estimation

Quality estimation is configured on the Task level.

In a new task

In an existing task

1. Go to the [task creation](#) page

4. Select one of the [validation modes](#) available

> Advanced configuration

▼ Quality

* Validation mode

NoneGround TruthHoneypots

* Frame selection method

Random per job

* Quantity per job

10%

Submit & OpenSubmit & Continue

- 5. Create the task and open the task page
- 6. Upload or create Ground Truth annotations in the Ground Truth job in the task
- 7. Switch the Ground Truth job into the `acceptance` stage and `completed` state

Job #12345	Ground truth	Assignee:	Stage:	State:	🗄️ Duration: 2 minutes	⋮
Created on November 25th 2024 18:54		Select a user	acceptance	new	❑ Frame count: 10 (83%)	
Last updated November 25th 2024 18:55			annotation			
			validation			
			acceptance			
			ground truth			
Job #12344	?	Assignee:	Stage:	State:	🗄️ Duration: 1 day	⋮
Created on July 8th 2024 23:46		Select a user		new	❑ Frame count: 12 (100%)	
Last updated November 25th 2024 18:56					🗄️ Frame range: 0-11	

A **Ground truth** job is considered **configured** if it is at the **acceptance** stage and in the **completed** state.

A *configured* Ground Truth job is required for all quality computations in CVAT.

Validation modes

Currently, there are 2 validation modes available for tasks: **Ground Truth** and **Honeypots**. These names are often used interchangeably, but in CVAT they have some differences. Both modes rely on the use of Ground Truth annotations in a task, stored in a [Ground Truth job](#), where they can be managed.

Ground Truth

In this mode some of the task frames are selected into the validation set, represented as a separate Ground Truth job. The regular annotation jobs in the task are not affected in any way.

Ground Truth jobs can be created at the task creation automatically or manually at any moment later. They can also be removed manually at any moment. This validation mode is available for any tasks and annotations.

This is a flexible mode that can be enabled or disabled at any moment without any disruptions to the annotation process.

Frame selection

This validation mode can use several frame selection methods.

Random

This is a simple method that selects frames into the validation set randomly, representing the [basic approach](#), described above.

Parameters:

- frame count - the number or percent of the task frames to be used for validation. Can be specified as an absolute number in the `Frame count` field or a percent in the `Quantity` field. If there are both fields on the page, they are linked, which means changing one of them will adjust the other one automatically.
- random seed - a number to be used to initialize the random number generator. Can be useful if you want to create a reproducible sequence of frames.

Random per job

This method selects frames into the validation set randomly from each annotation job in the task.

It solves one of the issues with the simple Random method that some of the jobs can get no validation frames, which makes it impossible to estimate quality in such jobs. Note that using this method can result in increased total size of the validation set.

Parameters:

- frame count per job - the percent of the job frames to be used for validation. This method uses segment size of the task to select the same number of validation frames in each job, if possible. Can be specified as an absolute number in the `Frame count` field or a percent in the `Quantity per job` field. If there are both fields on the page, they are linked, which means changing one of them will adjust the other one automatically.
- random seed - a number to be used to initialize the random number generator. Can be useful if you want to create a reproducible sequence of frames.

Honeypots

In this mode some random frames of the task are selected into the validation set. Then, validation frames are randomly mixed into regular annotation jobs. This mode can also be called “Ground Truth pool”, reflecting the way validation frames are used. This mode can only be used at task creation and cannot be changed later.

The mode has some limitations on the compatible tasks:

- It's not possible to use it for an already existing task, the task has to be recreated.
- This mode assumes random frame ordering, so it is only available for image annotation tasks and not for ordered sequences like videos.
- Tracks are not supported in such tasks.

The validation set can be managed after the task is created - annotations can be edited, frames can be excluded and restored, and honeypot frames in the regular jobs can be changed. However, it's not possible to select new validation frames after the task is created. The Ground truth job created

- frame count per job (%) - the percent of job frames (segment size) to be **added** into each annotation job from the validation set. Can be specified in the `overhead per job` field.
- total frame count (%) - the percent of the task frames to be included into the validation set. This value must result in at least `frame count per job * segment size` frames. Can be specified in the `Total honeypots` field.

Mode summary

Here is a brief comparison of the validation modes:

Aspect	Ground Truth	Honeypots
When can be used	any time	at task creation only
Frame management options	exclude, restore	exclude, restore, change honeypots in jobs
Ground Truth job management options	create, delete	create
Task frame requirements	-	random ordering only
Annotations	any	tracks are not supported
Minimum validation frames count	- <code>manual</code> and <code>random_uniform</code> - any (but some jobs can get no validation frames) - <code>random_per_job</code> - jobs count * GT frames per job	not less than honeypots count per job
Task annotation import	GT annotations and regular annotations do not affect each other	Annotations are imported both into the GT job and regular jobs. Annotations for validation frames are copied into corresponding honeypot frames.
Task annotation export	GT annotations and regular annotations do not affect each other	Annotations for non-validation frames are exported as is. Annotations for validation frames are taken from the GT frames. Honeypot frames are skipped.

Choosing the right mode

Here are some examples on how to choose between these options. The general advice is to use Ground Truth for better flexibility, but keep in mind that it can require more resources for validation set annotation. Honeypots, on the other hand, can be beneficial if you want to minimize the number of validation images required, but the downside here is that there are some limitations on where this mode can be used.

Example: a video annotation with tracks. In this case there is only 1 option - the Ground Truth mode, so just use it.

Example: an image dataset annotation, image order is not important. Here you can use both options. You can choose Ground Truth for better flexibility in validation. This way, you will have the full control of validation frames in the task, annotation options won't be limited, and the regular jobs will not be affected in any way. However, if you have a limited budget for the validation (for instance, you have only a small number of validation frames) or you want to allow more scalability (with this approach the number of validation frames doesn't depend on the number of regular annotation jobs), it makes sense to consider using Honeypots instead.

Quality management

If a task has a validation configured, there are several options to manage validation set images. With any of the validation modes, there will be a special Ground Truth (GT) job in the task.

Validation set management

Validation frames can be managed on the task Quality Management page. Here it's possible to check the number of validation frames, current validation mode and review the frame details. For each frame you can see the number of uses in the task. When in the Ground Truth mode, this number will be 1 for all frames. With Honeypots, these numbers can be 0, 1 or more.

In both validation modes it's possible to exclude some of the validation frames from being used for validation. This can be useful if you find that some of the validation frames are “bad”, extra, or if they have incorrect annotations, which you don't want to fix. Once a frame is marked “excluded”, it will not be used for validation. There is also an option to restore a previously excluded frame if you decide so.

There is an option to exclude or restore frames in bulk mode. To use it, select the frames needed using checkboxes, and click one of the buttons next to the table header.

Ground Truth job management

In the Ground Truth validation mode, there will be an option to remove the [Ground Truth job](#) from the task. It can be useful if you want to change validation set frames completely, add more frames, or remove some of the frames for any reason. This is available in the job Actions menu.

In the Honeypots mode, it's not possible to add or remove the GT job, so it's not possible to add more validation frames.

The screenshot displays the CVAT interface for a task named "Honeypot". The task details section shows a description, an issue tracker, and a "cat" label. The jobs list section shows two jobs: "Job #239180" (Ground truth, Completed) and "Job #232739" (In progress). The "Delete" button is highlighted in the actions menu.

Create

A Ground Truth job can be [added manually](#) in a task without a selected validation mode or in a task with the Ground Truth validation mode, after the existing Ground Truth job is [deleted manually](#).

Delete

To delete the Ground Truth job, do the following:

1. Open the task and find the Ground Truth job in the jobs list.
2. Click on three dots to open the menu.
3. From the menu, select **Delete**.

Note: The Ground truth job in the “Honeypots” task validation mode cannot be deleted.

1. Open the task and find the Ground truth job in the jobs list.
2. Click on three dots to open the menu.
3. From the menu, select **Import annotations**.
4. Select import format and select file.
5. Click **OK**.

Note that if there are imported annotations for the frames that exist in the task, but are not included in the **Ground truth** job, they will be ignored. This way, you don't need to worry about "cleaning up" your Ground truth annotations for the whole dataset before importing them. Importing annotations for the frames that are not known in the task still raises errors.

Export annotations

To export annotations from the Ground Truth job, do the following:

1. Open the task and find a job in the jobs list.
2. Click on three dots to open the menu.
3. From the menu, select **Export annotations**.

Annotation management

Annotations for validation frames can be displayed and edited in a special [Ground Truth job](#) in the task. You can edit the annotations manually, use auto-annotation features, import and export annotations in this job.

In the Ground Truth task validation mode, annotations of the ground Truth job do not affect other jobs in any way. The Ground Truth job is just a separate job, which can only be changed directly. Annotations from **Ground truth** jobs are not included in the dataset export, they also cannot be imported during task annotations import or with automatic annotation for the task.

In the Honeypots task validation mode, the annotations of the GT job also do not affect other jobs in any way. However, import and export of **task** annotations works differently. When importing **task** annotations, annotations for validation frames will be copied both into GT job frames and into corresponding honeypot frames in annotation jobs. When exporting **task** annotations, honeypot frames in annotation jobs will be ignored, and validation frames in the resulting dataset will get annotations from the GT job.

Note that it means that exporting from a task with honeypots and importing the results back will result in changed annotations on the honeypot frames. If you want to backup annotations, use a task backup or export job annotations instead.

Import and export of Ground Truth **job** annotations works the same way in both modes.

Ground Truth jobs are included in task backups, so can be saved and restored this way.

Import, Export, and Delete options are available from the Ground Truth job Actions menu. [Read more](#).

Annotation quality settings

If you need to tweak some aspects of comparisons, you can do this from the **Annotation Quality Settings** menu.

You can configure what overlap should be considered low or how annotations must be compared.

The updated settings will take effect on the next quality update.

To open **Annotation Quality Settings**, find **Quality report** and on the right side of it, click on three dots.

The following window will open. Hover over the **?** marks to understand what each field represents.

* Min overlap threshold (%)

* Low overlap threshold (%)

☒ Compare attributes

Keypoint Comparison 

* OKS sigma

Line Comparison 

* Relative thickness (frame side %)

☒ Check orientation

* Min similarity gain (%)

Group Comparison 

☒ Compare groups

* Min group match threshold (%)

Segmentation Comparison 

☒ Check object visibility

* Min visibility threshold (area %)

☒ Match only visible parts

Annotation quality settings have the following parameters:

Parameter	Description
<i>General reporting</i>	
Target metric	The primary metric used for quality estimation. It affects which metric is displayed in the UI and used for overall quality estimation.
<i>Immediate feedback</i>	
Max validations per job	Configures maximum job validations per assignment for the Immediate feedback feature.
Target metric threshold	Defines the minimal quality requirements in terms of the selected target metric. Serves as an acceptance threshold for the Immediate feedback feature.
<i>Shape matching</i>	
Min overlap threshold	Min overlap threshold used for the distinction between matched and unmatched shapes. Used to match all types of annotations. It corresponds to the Intersection over union (IoU) for spatial annotations, such as bounding boxes and masks.
Low overlap threshold	Low overlap threshold used for the distinction between strong and weak matches. Only affects <i>Low overlap</i> warnings. It's supposed that <i>Min similarity threshold</i> <= <i>Low overlap threshold</i> .

Empty names are annotated	Consider names annotated as "empty" if there are no annotations on a name. If a name is empty in both GT and job annotations, it will be considered a matching annotation.
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Point and Skeleton matching

OKS Sigma	Relative size of points. The percent of the bbox side, used as the radius of the circle around the GT point, where the checked point is expected to be. For boxes with different width and height, the "side" is computed as a geometric mean of the width and height.
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Point matching

Point size base	When comparing point annotations (including both separate points and point groups), the OKS sigma parameter defines a matching area for each GT point based on the object size. The point size base parameter allows configuring how to determine the object size. If set to <i>image_size</i> , the image size is used. Useful if each point annotation represents a separate object or boxes grouped with points do not represent object boundaries. If set to <i>group_bbox_size</i> , the object size is based on the point group bounding box size. Useful if each point group represents an object or there is a bbox grouped with points, representing the object size.
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Polyline matching

Relative thickness	Thickness of polylines, relative to the (image area) ^ 0.5. The distance to the boundary around the GT line inside of which the checked line points should be.
Check orientation	Indicates that polylines have direction. Used to produce <i>Mismatching direction</i> warnings
Min similarity gain (%)	The minimal gain in IoU between the given and reversed line directions to consider the line inverted. Only useful with the <i>Check orientation</i> parameter.

Group matching

Compare groups	Enables or disables annotation group checks. This check will produce <i>Group mismatch</i> warnings for grouped annotations, if the annotation groups do not match with the specified threshold. Each annotation within a group is expected to match with a corresponding annotation in a GT group.
Min group match threshold	Minimal IoU for groups to be considered matching, used when <i>Compare groups</i> is enabled.

Mask and polygon matching

Check object visibility	Check for partially-covered annotations. Masks and polygons will be compared to each other.
Min visibility threshold	Minimal visible area percent of the mask annotations (polygons, masks). Used for reporting <i>Covered annotation</i> warnings, useful with the <i>Check object visibility</i> option.
Match only visible parts	Use only the visible part of the masks and polygons in comparisons.

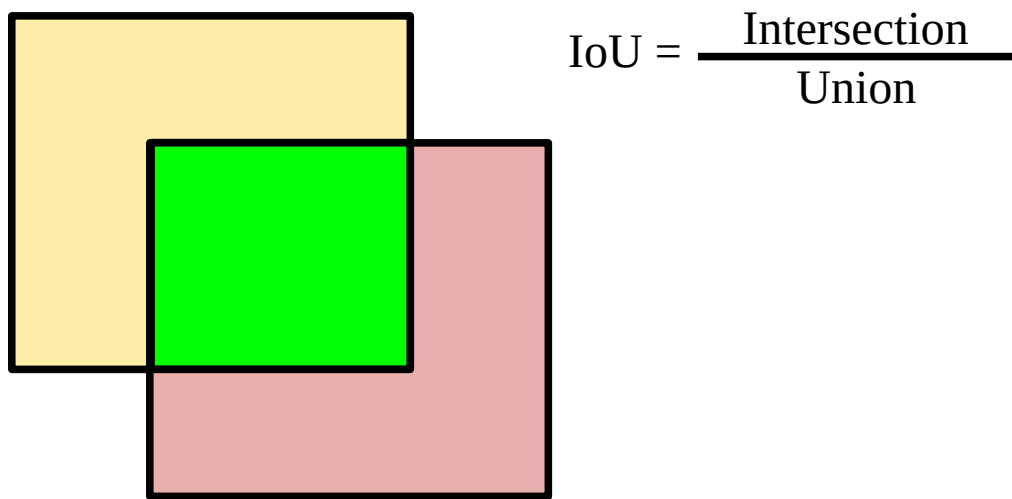
Comparisons

Tags

The equality is used for matching.

Shapes

A pair of shapes is considered matching, if both their shapes and labels match. For each shape, spatial parameters are matched first, then labels are matched.

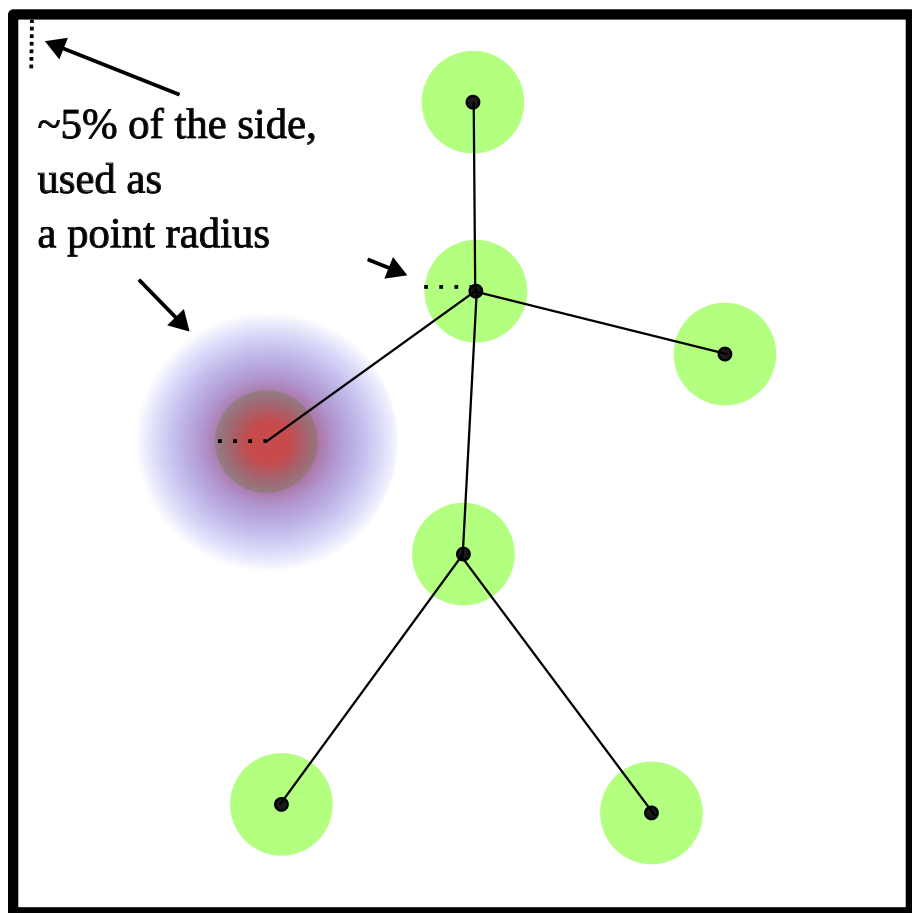


`IoU = intersection area / union area .`

The green part is the intersection, and green, yellow and red ones together are the union.

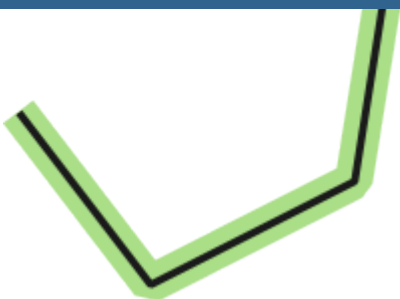
- polygons, masks - IoU. Polygons and masks are considered interchangeable, which means a mask can be matched with a polygon and vice versa. Polygons and masks in groups are merged into a single object first. If the [Match only visible parts](#) option is enabled, objects will be cut to only the visible (non-covered) parts only, which is determined by the shape z order.
- skeletons - The OKS metric [from the COCO](#) dataset is used. Briefly, each skeleton point gets a circular area around, determined by the *object size* (bounding box side) and *relative point size* (*sigma*) values, where this point can be matched with the specified probability. If a bounding box is grouped with the skeleton, it is used for object size computation, otherwise a bounding box of visible points of the skeleton is used.

For example, consider a skeleton with 6 points and a square bounding box attached:



In this example, the *Sigma* parameter is `0.05` (5%) of the bbox side. Areas shown in the green color cover ~68.2% (1 sigma) of the points, corresponding to each GT point. A point on the boundary of such an area will have ~88% of probability to be correct. The blue-colored zone contains ~95% (2 sigma) of the correct points for the corresponding GT point. A point on the boundary of such an area will have ~60% of probability to be correct. These probabilities are then averaged over the visible points of the skeleton, and the resulting values are compared against the *Min similarity threshold* to determine whether the skeletons are matching. *Sigma* corresponds to one from the [normal distribution](#).

- points - The OKS metric is used for each point group annotation. Same as for skeletons, *OKS Sigma* determines relative point sizes. The *Point size base* setting allows configuring whether points in point groups should use the group bounding box or the image space. Using image space for object size can be useful if you want to treat each point as a separate annotation.
- polylines - A pair of lines is considered matching if all the points of one line lie within a “hull” of the other one. The “hull” is determined as the area around the polyline, such as if the line had some “thickness”. For example, the black polyline can have a hull shown in the green color:



The line thickness can be configured via the *Relative thickness* setting. The value is relative to the image side and determines a half of the hull width.

- ellipses - IoU, described in more detail above.

Note: 2d cuboids are not supported

Tracks

Tracks are split into separate shapes and compared on the per-frame basis with other tracks and shapes.

Quality Analytics

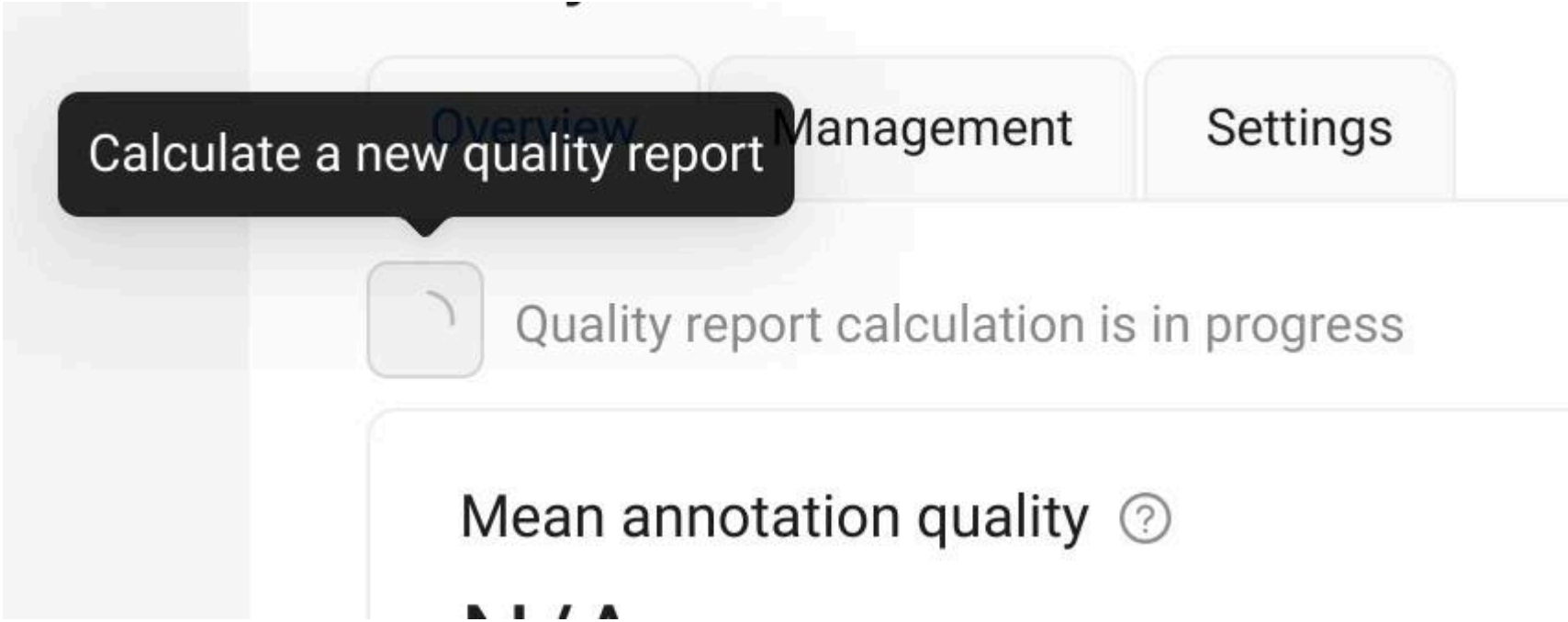
Note: quality analytics is a paid feature. Please check how to get access to this functionality in the [Paid features](#) section of the site.

Once the quality estimation is [enabled in a task](#) and the Ground Truth job is configured, quality analytics becomes available for the task and its jobs.

When you open the Quality Analytics page, it displays quality metrics from the most recent quality estimation. If it's your first time accessing the page, no quality report will be available yet. The date of the last computation is shown next to the report download button.

If you want to request updating of quality metrics in a task (e.g. after the settings were changed), you can do this by pressing the **Refresh** button on the task **Quality Management** > **Analytics** page.

Note that the process of quality calculation may take up to several hours, depending on the amount of data and labeled objects, and is **not updated immediately** after task updates.



Once quality metrics are computed, they are available for detailed review on this page. Conflicts can be reviewed in the [Review mode of jobs](#). A job must have at least 1 validation frame (shown in the **Frame intersection** column) to be included in quality computation.

Analytics page contents

The Analytics page has the following elements:

OverviewManagementSettings

C

Created 12 hours ago

Download

Mean annotation quality

90.2%

GT Conflicts

2

Errors: 1 (50.0%), Warnings: 1 (50.0%)

Issues

0

Resolved: 0

Job #574

Ground truth

Created on January 14th 2025 12:47

Last updated January 14th 2025 12:50

Assignee:

Select a user

Stage:

acceptance

State:

completed

Duration: 15 hours

Frame count: 150 (32%)

Jobs

Job	Stage	Assignee	Frame intersection	Conflicts	Quality	Download
Job #573	annotation		150 (32.4%)	2	90.2%	

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10 / page

Field	Description
Mean annotation quality	Displays the average quality of annotations, which includes: counts of the accurate annotations, total task annotations, ground truth annotations, accuracy, precision, and recall. The currently selected <i>Target metric</i> is displayed as the primary score.
GT Conflicts	Conflicts identified during quality assessment, including extra or missing annotations. Mouse over the ? icon for a detailed conflict report on your dataset.
Issues	Number of opened issues . If no issues were reported, 0 will be shown.
Quality report	Quality report in JSON format.
Ground truth job data	Information about ground truth job, including date, time, and number of issues.
List of jobs	List of all the jobs in the task

Job	Stage	Assignee	Frame intersection	Conflicts
Job #1359975	annotation	Alice	1 (25.0%)	3

Job #1359974	annotation	Bob	1 (25.0%)	5
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Problem Reporting

CVAT reports 2 possible error types: errors and warnings. Errors affect the resulting quality scores and highlight significant problems in annotations. Warnings do not affect the resulting quality metrics, but they still can highlight significant problems, depending on the project requirements.

Problem	Type	Description
Missing annotation	error	No matching annotation found in the regular job annotations. Configured by <i>Min overlap threshold</i> and shape type-specific parameters.

Extra annotation	error	No matching annotation found in the GT job annotations. Configured by <i>min overlap threshold</i> and shape type specific parameters.
Mismatching label	error	A GT and a regular job annotations match, but their labels are different.
Low overlap	warning	A GT and a regular job annotations match, but the similarity is low. Configured by <i>Low overlap threshold</i> .
Mismatching direction	warning	A GT and a regular lines match, but the lines have different direction. Configured by <i>Compare orientation</i> .
Mismatching attributes	warning	A GT and a regular annotations match, but their attributes are different. Configured by <i>Compare attributes</i> .
Mismatching groups	warning	A GT and a regular annotation groups do not match. Configured by <i>Compare groups</i> .
Covered annotation	warning	The visible part of a regular mask or polygon annotation is too small. The visibility is determined by arranging mask and polygon shapes on the frame in the specified <i>z order</i> . Configured by <i>Check object visibility</i> .

Quality Reports

For each job included in quality computation there is a quality report downloading button on the [Analytics page](#). There is also a button to download the aggregated task quality report. These buttons provide an option to download a Quality Report for a task or job in JSON format. Such reports can be useful if you want to process quality reported by CVAT automatically in your scripts etc.

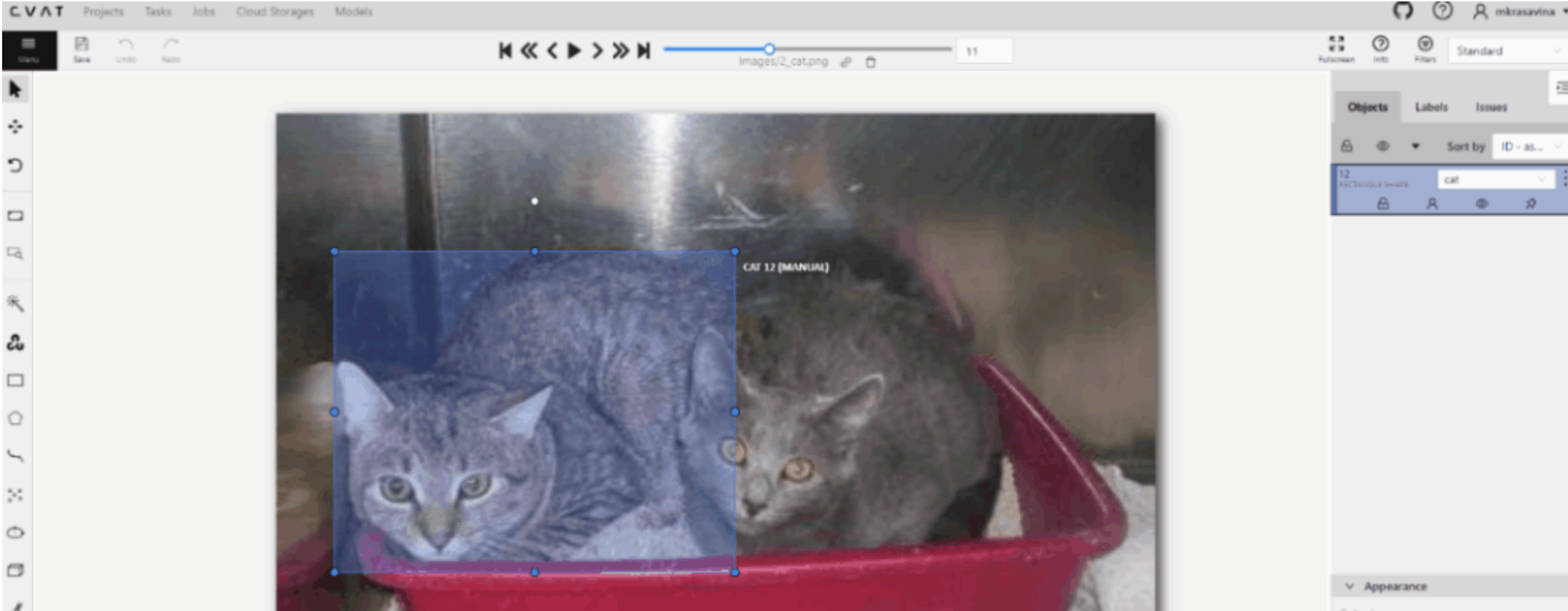
[Download](#)

Quality Reports contain quality metrics and conflicts, and include all the information available on the quality analytics page. You can find additional quality metrics in these reports, such as *mean_iou* for shapes, confusion matrices, per-label and per-frame quality estimations.

Additional information on how to compute and use various metrics for dataset quality estimation is available [here](#).

Reviewing GT conflicts

To see GT Conflicts in the CVAT interface, go to **Review > Issues > Show ground truth annotations and conflicts**.



Ground Truth annotations are displayed with a dotted-line border. The associated label and the (Ground Truth) marker are shown on hovering.

Upon hovering over an issue on the right-side panel with your mouse, the corresponding annotations are highlighted.

Use arrows in the Issue toolbar to move between GT conflicts.

To create an issue related to the conflict, right-click on the bounding box and from the menu select the type of issue you want to create.



Annotation quality & Honeypot video tutorial

This video demonstrates the process:

CVAT Product Tour #16: Annotation quality control with Ho...

